MIL-W-40145B

27 March 1969

SUPERSEDING
MIL-W-40145A
12 November 1965

MILITARY SPECIFICATION

WRENCHES, CROWFOOT, SPECIAL

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers one type of crowfoot wrench.

1.2 Classification.- Wrenches shall be of the following sizes, as specified (see 6.2).

Size - 3/4 inch
Size - 7/8 inch

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-P-416	-	Plating, Cadmium (Electrodeposited)
QQ-S-634	-	Steel, Bar, Carbon, Cold Finished, (Standard Quality)
QQ-Z-325	-	Zinc Coating, Electrodeposited, Requirements For

MILITARY

MIL-H-15424	-	Hand Tools, Packaging Of

STANDARD

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes

FSC 5120

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DRAWING

U. S. ARMY NATICK LABORATORIES

12-1-17 - Wrench, Crowfoot, Special

(Miniature copy of Drawing 12-1-17, identified as figure 1, is attached for information purposes only.)

(Copies of specifications, standards, and drawing required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

E 18-65 - Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race St., Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

* 3.1 First article.- When specified (see 6.2), before production is commenced, a sample wrench covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.2 Material.-

* 3.2.1 Steel bar.- The steel bar used for forging wrenches shall be of any commercial steel composition between 1035 to 1045 inclusive, in accordance with QQ-S-634, or any commercial alloy capable of being heat treated to the hardness specified in 3.3.2.

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3.3 Construction.-

* 3.3.1 Forming.- The wrenches shall be formed to the dimensions shown on Drawing 12-1-17 and in accordance with the manufacturer's current commercial practice.

* 3.3.2 Hardening.- The wrenches shall be heat treated to a hardness of Rockwell C40 to C45 when tested as specified in 4.4.1.

3.4 Torque performance.- The wrenches shall be capable of withstanding a torque of 720 inch-pounds for the 3/4 inch wrench and a torque of 1200 inch-pounds for the 7/8 inch wrench without deformation when tested as specified in 4.4.2.

3.5 Finish.- The finished wrenches shall be cadmium plated in accordance with type II, class 2 of QQ-P-416, or zinc plated in accordance with type II, class 2 of QQ-Z-325.

3.6 Marking for identification.- The wrenches shall be permanently marked with the manufacturer's name or trademark of such known characters as to be easily identifiable.

3.7 Workmanship.- The finished wrenches shall be clean and free from sharp or rough edges, burrs, pits, cracks, rust, and scale.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 First article inspection.- When a preproduction sample is required, it shall be examined for the defects listed in table I, dimensions specified, and tested as specified in 4.4.1 and 4.4.2.

* 4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

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4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.2 End item inspection.- The inspection lot shall consist of all wrenches of one size offered for inspection at one time. The sample unit shall be one complete wrench.

* 4.3.2.1 Visual examination.- Examination shall be made of the end item for defects in table I. The inspection level shall be level II with an acceptable quality level (AQL) of 4.0, expressed in terms of defects per hundred units.

TABLE I.- Classification of defects

Examine	Defect
Finish (plating)	Not zinc or cadmium plating Rust or scale on surfaces Area of no plating or non-adherent
Construction and workmanship	Fractured, broken, cracked or malformed Sharp or rough edges, burrs or projections
Identification marking	Missing, illegible, incorrect or incomplete

* 4.3.2.2 Dimensional examination.- Examination shall be made of the wrenches for compliance with dimensions specified. Any dimension not within specified tolerances shall constitute a defect. The inspection level shall be S-2 with an AQL of 4.0 defects per hundred units.

4.3.2.3 Testing of the end item.- Each sample wrench shall be tested in accordance with 4.4.1 and 4.4.2. The inspection level shall be S-2 with an AQL of 4.0 defects per hundred units.

4.3.2.4 Examination of preparation for delivery.- An examination shall be made to determine that preservation, packaging, packing, and marking as specified in MIL-H-15424 are complied with.

4.4 Tests.-

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4.4.1 Hardness test.- Hardness readings shall be made at both ends and in the center of the wrench in accordance with ASTM E 18-65. The average of the values obtained shall be evaluated to determine compliance with the requirements of 3.3.2. Any noncompliance shall constitute a defect.

4.4.2 Torque test.- The wrenches shall be tested on adaptors of the applicable size required to fit the wrench. The wrench shall be placed on the adaptor attached to a torque tester. Force shall be applied to the wrench to transmit the required test load between the wrench and the restrained adaptor. This shall be done a total of three times on each adaptor, rotating the wrench 30° to the next gripping position of the wrench after each torque cycle to determine compliance with 3.4. If the wrench slips on the adaptor before the test load values can be met, the wrench shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing.- Preservation, packaging, and packing shall be in accordance with the requirements for level A, B, or C of MIL-H-15424, as specified (see 6.2), and marking shall be as specified therein.

6. NOTES

6.1 Intended use.- The crowfoot wrenches covered by this specification are intended to be used for the purpose of installing and removing the ignition wiring harness nuts on internal combustion engines.

* 6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Size wrench required (see 1.2).
(c) First article.- When a preproduction sample is required (see 3.1).
(d) Selection of applicable level of preservation and packaging, and packing (see 5.1).

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6.3 The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL

Air Force - 84

Review activities:

Army - ME, WC

Air Force - 84, 85

User activities:

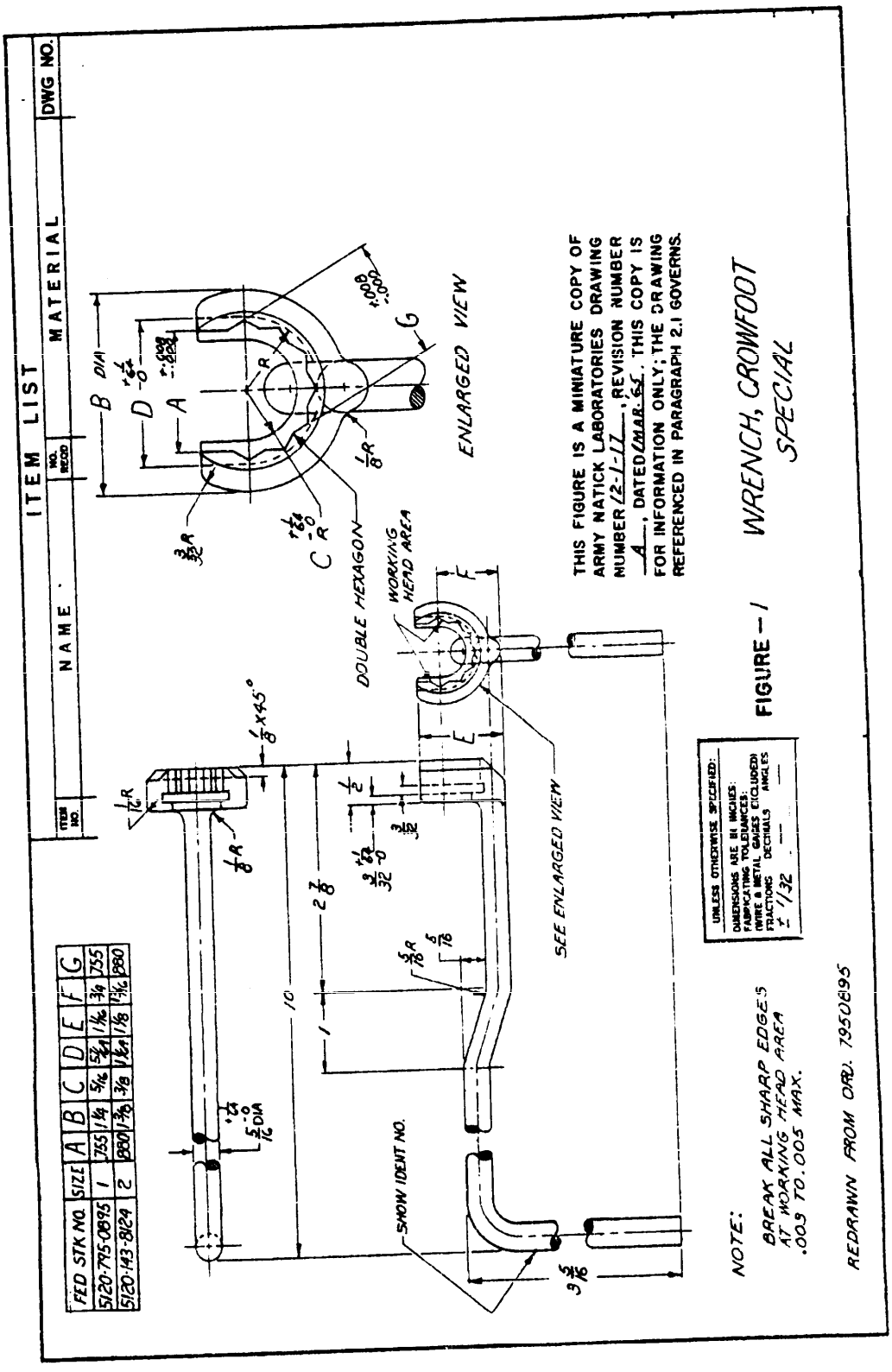
Army - AV, MI

Preparing activity:

Army - GL

Project No. 5120-0644

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Wrenches, Crowfoot, Special MIL-W-40145B

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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